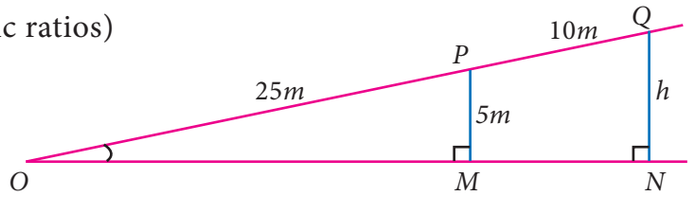
10. A boy standing at a point $O$ finds his kite flying at a point $P$ with distance $OP=25m$. It is at a height of $5m$ from the ground. When the thread is extended by $10m$ from $P$, it reaches a point $Q$. What will be the height $QN$ of the kite from the ground? (use trigonometric ratios)




## 6.2 Trigonometric Ratios of Some Special Angles

The values of trigonometric ratios of certain angles can be obtained geometrically. Two special triangles come to help here.

### 6.2.1 Trigonometric ratios of 45°

Consider a triangle $ABC$ with angles 45°, 45° and 90° as shown in the figure 6.9.

It is the shape of half a square, cut along the square's diagonal. Note that it is also an isosceles triangle (both legs have the same length, $a$ units).

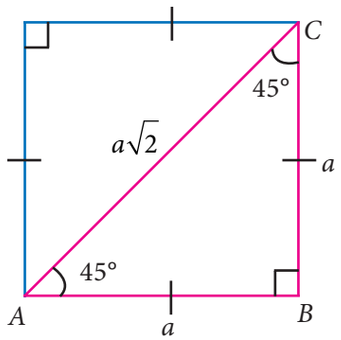


Fig. 6.9

Use Pythagoras theorem to check if the diagonal is of length $a\sqrt{2}$.

Now, from the right-angled triangle $ABC$,

$$\sin 45^o = \frac{opposite\ side}{hypotenuse} = \frac{BC}{AC} = \frac{a}{a\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$\cos 45^o = \frac{adjacent\ side}{hypotenuse} = \frac{AB}{AC} = \frac{a}{a\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$\tan 45^o = \frac{opposite\ side}{adjacent\ side} = \frac{BC}{AB} = \frac{a}{a} = 1$$

The reciprocals of these ratio can be easily foundout to be

$\operatorname{cosec} 45° = \sqrt{2}$ ;

$\sec 45° = \sqrt{2}$ and

$\cot 45° = 1$

### 6.2.2 Trigonometric Ratios of 30° and 60°

Consider an equilateral triangle $PQR$ of side length 2 units.

Draw a bisector of $\angle P$. Let it meet $QR$ at $M$.

$PQ = QR = RP = 2$ units.

$QM = MR = 1$ unit (Why?)

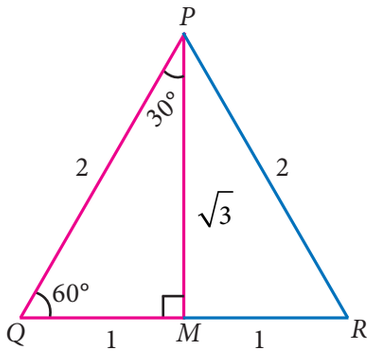


Fig. 6.10

Knowing $PQ$ and $QM$, we can find $PM$, using Pythagoras theorem,



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